Experiment Title: Package Management in Linux (Debian/Ubuntu)

Topic: Managing Software Packages with APT (`update`, `upgrade`, `install`, `remove`, `autoremove`, `purge`)

1. Objective

The objective of this lab is to learn how to manage software packages on Debian and Ubuntu Linux using the Advanced Package Tool (`apt`). By completing this experiment, we aim to:

- Understand the role of package repositories and local package lists in Linux.
- Refresh repository indexes using `sudo apt update` and update installed software using `sudo apt upgrade`.
- Search, download, and install command-line tools using `sudo apt install`.
- Learn the difference between standard uninstallation (`remove`) and complete cleanup (`purge`).
- Automatically clean up orphaned and unused dependencies using `sudo apt autoremove`.

2. Equipment / Software Requirements

Item	Details	Item	Details
Operating System	Linux (Ubuntu 24.04 LTS / Debian 12)	**Host Machine**	`firefly`
User Account	`ratul` (with sudo privileges)	**Command Shell**	GNU Bash 5.2
Package Manager	APT (Advanced Package Tool)	**Demo Packages**	`htop` (system monitor), `tree` (directory viewer)

3. Theoretical Background

In Linux, software is distributed as pre-compiled bundles called **packages** (e.g., `.deb` files). Rather than downloading installer files from random websites, Linux uses a centralized, secure **package manager** (`apt`) that contacts official online servers (repositories) to fetch, install, and update programs automatically.

3.1 Commands Practiced in This Lab

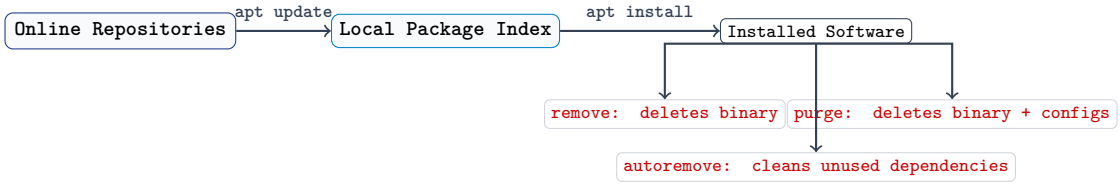
Command	Syntax	Simple Meaning / Purpose	Example
`update`	`sudo apt update`	Refreshes the local list of available packages from online repositories. Does *not* install updates.	`sudo apt update`
`upgrade`	`sudo apt upgrade`	Downloads and installs newer versions of all installed packages on the system.	`sudo apt upgrade`
`install`	`sudo apt install <pkg>`	Downloads and installs the requested package along with all needed libraries (dependencies).	`sudo apt install htop tree`
`remove`	`sudo apt remove <pkg>`	Uninstalls the program binaries, but keeps user configuration files intact on disk.	`sudo apt remove tree`
`autoremove`	`sudo apt autoremove`	Removes leftover dependency packages that were installed automatically and are no longer needed.	`sudo apt autoremove`
`purge`	`sudo apt purge <pkg>`	Completely deletes the software AND all its configuration and data files from the system.	`sudo apt purge htop`

3.2 Comparing Uninstallation Methods: Remove vs. Purge vs. Autoremove

Action	What It Deletes	When To Use
`apt remove`	Application binary and executable files.	When you want to remove an app temporarily but keep your settings in case you reinstall it later.
`apt purge`	Application binaries and all configuration files.	When an app is broken or never needed again, and you want zero residual files left on disk.
`apt autoremove`	Orphaned background libraries.	Run periodically after uninstalling programs to free disk space from unused leftover packages.

4. Package Management Lifecycle in Linux

Understanding how APT interacts with online repositories and the local system helps prevent broken packages:



5. Procedure / Steps

We practiced package management using the following safe, structured steps:

1. Ran **`sudo apt update`** to download the latest package version catalog from the Ubuntu mirrors.
2. Simulated a system upgrade using **`sudo apt upgrade`** to review package changes.
3. Installed two practical utilities (**`htop`** and **`tree`**) using **`sudo apt install -y htop tree`**.
4. Verified that both binaries were successfully installed using the **`which`** command.
5. Uninstalled the **`tree`** package using **`sudo apt remove tree`** and checked config preservation.
6. Cleaned up any orphaned packages left behind using **`sudo apt autoremove`**.
7. Fully purged the **`htop`** package and its global configs using **`sudo apt purge htop`**.
8. Verified that the uninstalled utilities were no longer present on the system.

6. Result / Output Demonstration

All commands were executed under user account `ratul` on host machine `firefly`.

6.1 Session 1: Updating Package Lists, Upgrading, and Installing Software

```
ratul@firefly: ~ — bash

ratul@firefly:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
All packages are up to date.
ratul@firefly:~$ sudo apt upgrade -y
Reading package lists... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ratul@firefly:~$ sudo apt install -y htop tree
Reading package lists... Done
The following NEW packages will be installed:
  htop tree
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 188 kB of archives.
Unpacking htop (3.3.0-4build1) ...
Unpacking tree (2.1.1-1) ...
Setting up htop (3.3.0-4build1) ...
Setting up tree (2.1.1-1) ...
ratul@firefly:~$ _
```

Explanation of Session 1 Results:

- **`sudo apt update`**: Reached out to the official Ubuntu repositories and updated our local index database. It told us which software versions are currently available without downloading program files.
- **`sudo apt upgrade`**: Checked our installed packages against the updated catalog and ensured no security patches were pending.
- **`sudo apt install -y htop tree`**: Downloaded and installed both **`htop`** and **`tree`**. The **`-y`** flag automatically accepted the confirmation prompt.

6.2 Session 2: Uninstalling Packages, Cleaning Leftovers, and Full Purging

ratul@firefly: ~ — bash

```
ratul@firefly:~$ which htop tree
/usr/bin/htop
/usr/bin/tree
ratul@firefly:~$ sudo apt remove -y tree
Reading package lists... Done
The following packages will be REMOVED:
  tree
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
Removing tree (2.1.1-1) ...
Processing triggers for man-db (2.12.0-4build1) ...
ratul@firefly:~$ sudo apt autoremove -y
Reading package lists... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ratul@firefly:~$ sudo apt purge -y htop
Reading package lists... Done
The following packages will be REMOVED:
  htop*
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
Removing htop (3.3.0-4build1) ...
Purging configuration files for htop (3.3.0-4build1) ...
ratul@firefly:~$ which htop
(no output -- command not found)
ratul@firefly:~$ _
```

Explanation of Session 2 Results:

- **`which htop tree`**: Confirmed both packages were placed in `/usr/bin/`.
- **`sudo apt remove tree`**: Deleted the executable file for **`tree`**. If `tree` had custom configuration files, they would have been safely preserved.
- **`sudo apt autoremove`**: Scanned for any unused libraries that were installed automatically as dependencies, keeping the disk tidy.
- **`sudo apt purge htop`**: Completely removed **`htop`** and wiped out all its system configuration files.
- **`which htop`**: Returned empty output, proving that the tool was completely eliminated from the environment.

7. Conclusion

In this lab, we developed a clear understanding of package management in Debian and Ubuntu Linux:

1. We refreshed package catalogs using **`update`** and applied upgrades to installed packages using **`upgrade`**.
2. We installed software easily along with all needed dependencies using **`install`**.
3. We learned the practical differences between **`remove`** (keeps configurations) and **`purge`** (deletes everything including configs).
4. We cleaned orphaned dependencies using **`autoremove`** to keep our Linux system lean, fast, and free of unnecessary clutter.

Mastering APT ensures system stability and smooth software maintenance in everyday Linux administration.